

PROPERTIES OF MATTER II (12)

Aim

To determine the coefficient of viscosity of water by the horizontal flow method.

Materials and Apparatus

Constant pressure head, capillary tube, piece of thread, two beakers, measuring cylinder, meter ruler, stopwatch, stands, spirit level, travelling microscope.

Theory

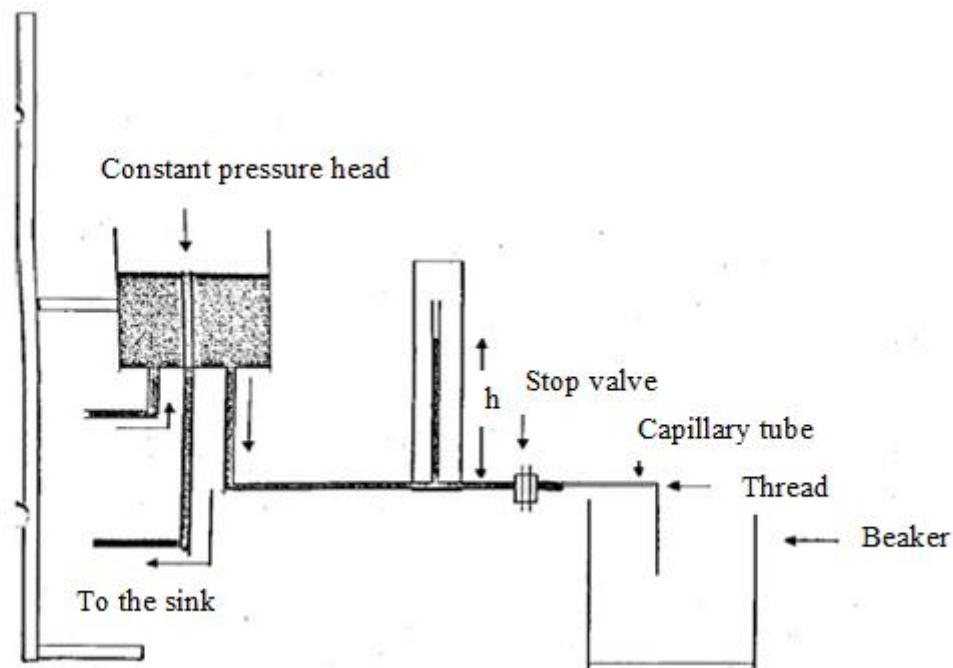


Fig. 1: Experimental setup to determine the viscosity of water

Where; h : height of the water column

Volume rate of flow $\frac{dV}{dt}$ of water through the horizontal capillary tube given by,

$$\frac{dV}{dt} = \frac{\pi a^4}{8\eta} \frac{dP}{dx} \quad (1)$$

Where,

$\frac{dV}{dt}$ - Volume flow rate of the liquid

a - radius of capillary tube

η - Coefficient of viscosity of water

$\frac{dP}{dx}$ - Pressure gradient

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By integrating,

$$\frac{V}{t} = \frac{\pi a^4}{8\eta} \frac{\Delta P}{\Delta x} \quad (2)$$

Where,

V – volume of the water collected in time t

ΔP - pressure difference across Δx distance

Δx – length of the capillary tube

But from hydrostatics,

$$\Delta P = h\rho g \quad (3)$$

Where,

h – height of water column

g – acceleration due to gravity

ρ – density of water

$$\frac{V}{t} = \frac{\pi a^4}{8\eta} \frac{\rho g}{\Delta x} h \quad (4)$$

The coefficient of viscosity can be found from the gradient of the graph of $\frac{V}{t}$ vs h .

Procedure

- a) Clean the capillary tube using the standard method.
- b) Arrange the setup as shown in Fig. 1 and level the capillary tube using a spirit level.

Q. Why should we level the capillary tube?

- c) Obtain a steady flow of water through the capillary tube by adjusting a constant pressure head and stop valve.

Q. Why do we require a steady flow?

- d) Collect water for a period of two minutes.
- e) Measure the corresponding height h and the volume of the water V .
- f) Repeat for another seven more values of h .
- g) Note down the room temperature.

Q. Why is it necessary to record the room temperature?

- h) Find the coefficient of viscosity of water by plotting a graph.